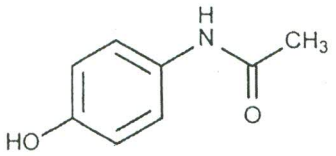
Analytical Chemistry Cumulative Exam
December 2, 2016

1. Describe how you could selectively extract acetaminophen from codeine using octanol and water at any pH value. Tylenol 3 is a mixture of these two drugs.

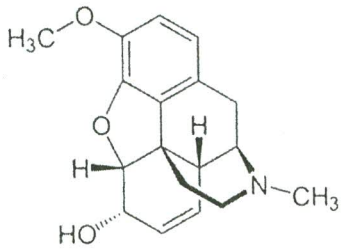
For this problem, define selectivity as greater than 50% of each component is in one phase or the other.

The *log P* for acetaminophen is 0.339, where *P* is the octanol/water partition coefficient. Octanol/water partition coefficients are for the neutral forms only. Acetaminophen is neutral (while the phenol is acidic, its pK_a is quite high ~ 10, assume it is neutral for this exercise).

For codeine, *log P* is 1.20 and the pK_a is 8.15 (NH^+)

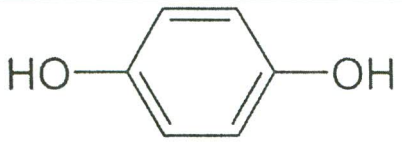
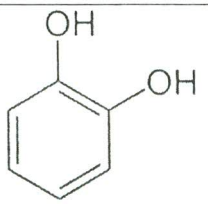


Acetaminophen



Codeine

2. The van Deemter equation is one of the most important and widely discussed equations in separations science. It gives much insight into separation processes. Provide the van Deemter equation (or the Knox equation which is an updated version for liquid-based separations) and define each variable and explain that individual variable's importance to separations.

3. Consider the separation of catechol and hydroquinone on a standard C18 column. Their *log P* values are listed below. Based on this information predict their retention behavior. Also include a description of the possible molecular mechanisms involved with this separation. Describe why it is possible to easily separate these compounds despite their similar structures and *log P* values.

Hydroquinone	**Catechol**
log P = 0.64	*log P* = 0.88